

Your First Queries

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Learning DQL Fundamentals

Dynatrace Query Language (DQL) is how you access data in Grail. This notebook introduces the core concepts and patterns you'll use daily.

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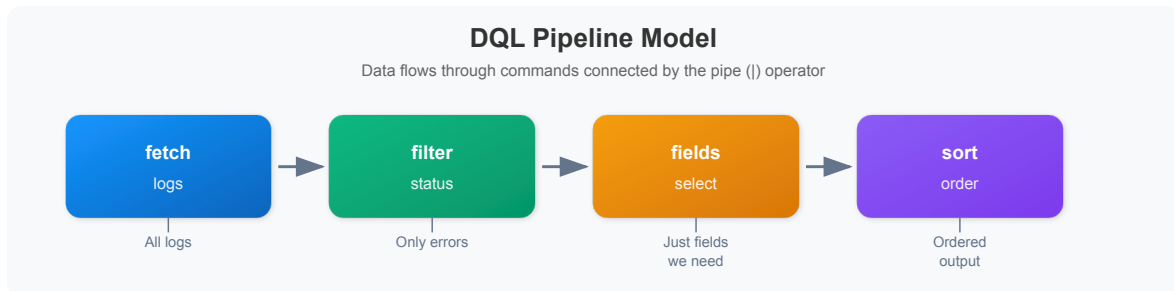
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Prerequisites

- Dynatrace environment with data (ONBRD-05, ONBRD-06, ONBRD-07)
- DQL query permissions
- Access to Notebooks or the DQL query interface

1. DQL Basics

DQL is a **pipeline-based query language**—not SQL. Data flows through a series of commands connected by the pipe (`|`) operator.



DQL vs SQL

DQL	SQL	Note
<code>fetch logs</code>	<code>SELECT * FROM logs</code>	Start with data source
<code>\ filter x == "y"</code>	<code>WHERE x = 'y'</code>	Use <code>==</code> , double quotes
<code>\ fields a, b</code>	<code>SELECT a, b</code>	Field selection after fetch
<code>\ summarize count()</code>	<code>SELECT COUNT(*)</code>	Aggregation command
<code>by: {field}</code>	<code>GROUP BY field</code>	Grouping syntax
<code>{"a", "b"}</code>	<code>('a', 'b')</code>	Array syntax (curly braces)

2. The Pipeline Model

Each command in the pipeline operates on the output of the previous command:

```
fetch logs                // 1. Get all logs
| filter loglevel == "error" // 2. Keep only errors
| filter timestamp > now() - 1h // 3. Last hour only
| fields timestamp, content // 4. Select columns
| sort timestamp desc      // 5. Order by time
| limit 100               // 6. Take first 100
```

Order Matters

- **Filter early** - Reduces data before expensive operations
- **Select fields** - Reduces memory usage
- **Aggregate** - Summarize before sorting
- **Sort** - Order the final results
- **Limit** - Control output size

3. Fetching Data

Every DQL query starts with a `fetch` command specifying the data source.

```
// Fetch logs (most common)
fetch logs, from:-1h
| limit 10
```

```
// Fetch spans (distributed traces)
fetch spans, from:-1h
| limit 10
```

```
// Fetch entity data (hosts)
fetch dt.entity.host
| limit 10

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | limit 10
```

```
// Fetch problems
fetch dt.davis.problems, from:-24h
| limit 10
```

Common Data Sources

Source	Description
<code>logs</code>	Log records

Source	Description
<code>spans</code>	Distributed trace spans
<code>events</code>	System events
<code>bizevents</code>	Business events
<code>dt.entity.host</code>	Host entities
<code>dt.entity.service</code>	Service entities
<code>dt.entity.process_group</code>	Process group entities
<code>dt.davis.problems</code>	Detected problems

4. Filtering

Use `filter` to narrow results. Filter as early as possible for performance.

```
// Filter by equality
fetch logs, from:-1h
| filter loglevel == "error"
| limit 20
```

```
// Filter with multiple conditions (AND)
fetch logs, from:-1h
| filter loglevel == "error"
| filter timestamp > now() - 1h
| limit 20
```

```
// Filter with or condition
fetch logs, from:-1h
| filter loglevel == "error" or loglevel == "warn"
| limit 20
```

```
// Filter using IN for multiple values
fetch logs, from:-1h
| filter in(loglevel, {"error", "warn", "fatal"})
| limit 20
```

```
// Filter with string matching
fetch logs, from:-1h
| filter contains(content, "timeout")
| limit 20
```

Filter Operators

Operator	Example	Description
<code>==</code>	<code>field == "value"</code>	Equals
<code>!=</code>	<code>field != "value"</code>	Not equals
<code>></code> , <code><</code>	<code>count > 10</code>	Greater/less than
<code>>=</code> , <code><=</code>	<code>count >= 10</code>	Greater/less or equal
<code>and</code> , <code>or</code>	<code>a == 1 and b == 2</code>	Logical operators
<code>in()</code>	<code>in(field, {"a", "b"})</code>	Value in set
<code>contains()</code>	<code>contains(field, "text")</code>	Substring match
<code>isNull()</code>	<code>isNull(field)</code>	Field is null
<code>isNotNull()</code>	<code>isNotNull(field)</code>	Field is not null

5. Selecting Fields

Use `fields` to select specific columns. This improves readability and performance.

```
// Select specific fields from logs
fetch logs, from:-1h
| fields timestamp, loglevel, log.source, content
| limit 20
```

```
// Create calculated fields
fetch spans, from:-1h
| fields span.name,
      duration,
      duration_ms = duration / 1000000.0
| limit 20
```

```
// Rename fields with aliases
fetch dt.entity.host
| fields name = entity.name,
        status = state,
        os = osType
| limit 20
```

fieldsAdd vs fields

Command	Effect
<code>fields</code>	Keeps only specified fields
<code>fieldsAdd</code>	Adds new fields, keeps all existing

```
// fieldsAdd keeps existing fields and adds new ones
fetch spans, from:-1h
| fieldsAdd duration_ms = duration / 1000000.0
| fields span.name, duration, duration_ms
| limit 10
```

6. Aggregating with Summarize

Use `summarize` to aggregate data. Combine with `by:` for grouping.

```
// Simple count
fetch logs, from: now() - 1h
| summarize total_logs = count()
```

```
// Count by group
fetch logs, from: now() - 1h
| summarize log_count = count(), by: {loglevel}
| sort log_count desc
```

```
// Multiple aggregations
fetch spans, from: now() - 1h
| filter span.kind == "server"
| summarize
    request_count = count(),
    avg_duration = avg(duration),
    max_duration = max(duration),
    by: {service.name}
| sort request_count desc
| limit 20
```

```
// Conditional counting
fetch spans, from: now() - 1h
| filter span.kind == "server"
| summarize
    total = count(),
    errors = countIf(span.status_code == "error"),
    by: {service.name}
| fieldsAdd error_rate = 100.0 * errors / total
| sort error_rate desc
| limit 20
```

Common Aggregation Functions

Function	Description
<code>count()</code>	Count records
<code>countIf(condition)</code>	Count where condition is true
<code>sum(field)</code>	Sum values
<code>avg(field)</code>	Average value
<code>min(field)</code>	Minimum value
<code>max(field)</code>	Maximum value
<code>percentile(field, 95)</code>	95th percentile

7. Sorting and Limiting

Control output order and size.

```
// Sort descending (newest first)
fetch logs, from: now() - 1h
| fields timestamp, loglevel, content
| sort timestamp desc
| limit 20
```

```
// Sort by multiple fields
fetch spans, from: now() - 1h
| filter span.kind == "server"
| summarize request_count = count(), by: {service.name}
| sort request_count desc
| limit 10
```

```
// Find slowest spans
fetch spans, from: now() - 1h
| fields span.name, service.name, duration
| sort duration desc
| limit 10
```

8. Time Ranges

Control the time range for your queries.

```
// Last getHour(using from: parameter)
fetch logs, from: now() - 1h
| summarize count()
```

```
// Specific time range
fetch logs, from: now() - 24h, to: now() - 12h
| summarize count()
```

```
// Last 7 days
fetch dt.davis.problems, from: now() - 7d
| summarize problem_count = count(), by: {status}
```


Time Units

Unit	Example	Description
s	<code>now() - 30s</code>	Seconds
m	<code>now() - 15m</code>	Minutes
h	<code>now() - 2h</code>	Hours
d	<code>now() - 7d</code>	Days

9. Common Patterns

Here are patterns you'll use frequently.

Error Investigation

```
// Find error logs with context
fetch logs, from: now() - 1h
| filter loglevel == "error"
| fields timestamp, log.source, content
| sort timestamp desc
| limit 50
```

Service Performance

```
// Service response time summary
fetch spans, from: now() - 1h
| filter span.kind == "server"
| summarize
  requests = count(),
  avg_ms = avg(duration) / 1000000.0,
  p95_ms = percentile(duration, 95) / 1000000.0,
  by: {service.name}
| sort requests desc
| limit 20
```

Log Volume Analysis

```
// Log volume by source and severity
fetch logs, from: now() - 1h
| summarize log_count = count(), by: {log.source, loglevel}
| sort log_count desc
| limit 30
```

Entity Inventory

```
// Host inventory with details
fetch dt.entity.host
| fields
    name = entity.name,
    state,
    os = osType,
    cores = cpuCores
| sort name
| limit 50
```

10. Next Steps

With DQL fundamentals covered:

1. **ONBRD-09: Setting Up Alerts** - Configure alerting and notifications
2. Practice with your own data
3. Explore the DQL documentation for advanced functions
4. Try creating queries in Notebooks

DQL Checklist

- ☐ Understand the pipeline model
- ☐ Can fetch different data types
- ☐ Can filter with conditions
- ☐ Can select and calculate fields
- ☐ Can aggregate with summarize
- ☐ Can sort and limit results

- [] Can specify time ranges
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Summary

In this notebook, you learned:

- DQL uses a pipeline model (not SQL)
 - `fetch` starts every query
 - `filter` narrows results
 - `fields` selects and calculates columns
 - `summarize` aggregates data
 - `sort` and `limit` control output
 - Time ranges with `from:` and `to:`
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References

- [DQL Reference](#)
 - [DQL Functions](#)
 - [DQL Commands](#)
 - [DQL Examples](#)
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